Q.25 Discuss various advantages of ribs.

Q.26 Discuss tolerance and its types.

Q.27 Discuss various types of holes used in plastic product design.

Q.28 Explain feasibility study and also name its various types.

Q.29 Explain undercuts and their types.

Q.30 Discuss various processing limitations of polymer product design.

Q.31 Explain hot tool welding with diagram.

Q.32 Explain friction or spin welding with diagram.

Q.33 Explain ribs and bosses (with neat sketch).

Q.34 Discuss drilling process for plastics.

Q.35 Explain various preliminary design considerations in plastic product design.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Discuss various stages of new product development.

Q.37 Explain:

a) Cost economics and its effect on plastic product design.

b) Product life cycle and its stages.

Q.38 Write short note on;

a) Various materials and their selection criteria.

b) Explain gate side and its location.

No. of Printed Pages : 4
Roll No. ....

182265/122265/ 032256/2263

6th Sem / Plastic, Chem. Engg. (Spl. Polymer Tech.)
Subject:- Plastic Product Design

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 _______ is not a texturing method:

a) Honning b) EDM texturing

c) Photochemical etching

d) Sand blast coating

Q.2 _______ should be the minimum meeting angle where weld line disappear.

a) 90-1000 b) 60-800

c) 120-1500 d) 180-2000

Q.3 Name the solvent used for PVC _______.

a) MEK b) MCL

c) Cyclo hexanone d) All of them

Q.4 _________ should be the draft angle used in plastic products.

a) 50 b) 60

c) 30 d) 1/20

Q.5 The aim of feasibility study is __________.

a) to determine sources and profitability of organisation

b) to improve mould design

c) to give reaction in polymers

d) none of the above

Q.6 Which the most difficult shape to prepare?

a) Artistic shapes b) Engineering shapes

c) Plain utility d) None of the above

Q.7 What must be avoided while designing moulded component in plastic product design?

a) Radii b) bend

c) fillet d) sharp corners

Q.8 Method of adhesion, in which parent material is used as dope is known as __________.

a) Solvent adhesion b) solvent cementing

c) Solvent doping d) none of the above

Q.9 Alphabets and letters are generally used for ______.

a) Identification of shift

b) for displaying company logos

c) For texturing effect d) All of these

Q.10 Which of the following is not a type of welding?

a) ultrasonic b) induction

c) hot gas d) rivet

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Name two main types holes used in plastic product design.

Q.12 MEK stands for _________.

Q.13 Give two advantages of Ribs.

Q.14 Name two permanent joining methods used in plastic assembly.

Q.15 Dope cement generally contains ________ % of parent resins.

Q.16 Line formed by meeting of two parallel flow fronts is known as _________.

Q.17 Give two disadvantages of non-uniform wall thickness.

Q.18 Alphabets are letters are generally used for _______.

Q.19 Name various types of shapes used in plastic product design.

Q.20 Define bosses?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain fillet and rounds in plastic product design.

Q.22 Discuss wall thickness and its importance in plastic product design.

Q.23 Write short note on surface finishing.

Q.24 Discuss various type of threads used in plastic product design.